

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)	(i)		volume is 50 cm ³ therefore mass is 50 g (1) density = 50/54.5 = 0.917 g cm ⁻³ (1)		2		2	1	
		(ii)		any of following for (1) - hydrogen bonding exists in both water and ice but it extends throughout the whole structure in ice - in ice hydrogen bonds hold the molecules together in an open (tetrahedral) structure - the molecules are further apart in ice than they are in water (so ice is less dense than water)	1			1		
				Question 6 total	9	8	3	20	9	0